

First Order Logic - Handout

1 The syntax

1.1 The alphabet

- (a) A countable (non-empty) set $\mathcal{V}$ of *variables* $x_1, x_2, \dots$
- (b) A countable (non-empty) set of *predicate symbols* $p_1^n, p_2^n, \dots$, for any $n(\in \mathbf{N})$, where n denotes the *arity* of the predicate symbol p_i^n
- (c) A countable (possibly empty) set of *function symbols* $f_1^n, f_2^n, \dots$, for any $n(\in \mathbf{N})$, where n denotes the arity of the function symbol f_i^n
- (d) A countable (possibly empty) set of *constant symbols* $c_1, c_2, \dots$
- (e) *Logical Connectives* $\neg, \rightarrow$
- (f) *Universal quantifier* $\forall$
- (g) Paratheses $(,)$

For a language to qualify as an FOL-language, its alphabet must therefore have, at least, variables, predicate symbols, the logical connectives, quantifier and paratheses. The interpretation of the symbols in the alphabet is done in a non-empty domain of discourse, the latter considered to be a set. Predicate symbols are intended to be interpreted as relations over the domain, function symbols as functions on the domain, and constant symbols as distinguished elements of the domain.

Note that countability of the sets in the alphabet is not a necessary assumption – an FOL-language may also involve uncountable sets in its alphabet.

1.2 Terms

Definition 1.1. The *set $\mathcal{T}$ of terms* is the smallest set of strings (finite sequences of symbols) over the sets of variables, constants and function symbols, such that

- (a) any variable is a term (i.e. a member of $\mathcal{T}$),
- (b) any constant is a term,
- (c) if f^n is a function symbol of arity n and $t_1, t_2, \dots, t_n$ are terms, $f^n t_1 t_2 \dots t_n$ is a term.

A term is said to be *closed*, if it involves constants only.

Example 1.1. Suppose the interpretation of an FOL-language is to be done in the domain of natural numbers. Then, instead of the abstract symbols mentioned in the alphabet, we may use familiar operator or relation symbols. So, for instance, the alphabet may contain apart from variables, binary predicate symbols $=, <$, binary function symbols $+, \times$, and the constant symbol 9.

Some examples of terms, apart from the variables and constant 9 would be: $x + z$, $(x + y) \times 9$, $y \times ((y \times y) + y)$, or $9 + 9$. The last one is a closed term, and so is 9.

1.3 Well-formed formulae (wffs)

Terms, by themselves, are not sentences. They are used to build wffs of FOL, the simplest of which involve predicate symbols and are called *atomic*. The definition of wffs, like that of terms, is recursive.

Definition 1.2. The *set $\mathcal{F}$ of wffs* is the smallest set of strings over the alphabet given above, such that

- (a) if p^n is a predicate symbol of arity n and $t_1, t_2, \dots, t_n$ are terms, $p^n t_1 t_2 \dots t_n$ is a wff, called an *atomic wff*,
- (b) if α is a wff, so is $\neg \alpha$,
- (c) if α, β are wffs, $\alpha \rightarrow \beta$ is also a wff,
- (d) if x is a variable and α is a wff, $\forall x \alpha$ is a wff.

As in PL, we have the usual abbreviations defining the connectives of conjunction ('and'), disjunction ('or') and bi-implication ('if and only if'). The new entrant in FOL is the *existential quantifier*, 'there exists', given in (d) below.

Abbreviations

- (a) $\alpha \wedge \beta := \neg(\alpha \rightarrow \neg\beta)$.
- (b) $\alpha \vee \beta := \neg\alpha \rightarrow \beta$.
- (c) $\alpha \leftrightarrow \beta := (\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$.
- (d) $\exists x\alpha := \neg\forall x(\neg\alpha)$.

px is an atomic wff, where p is a unary predicate symbol. Some other wffs: $px \rightarrow px$, $\forall xpx$, $\exists xpx \wedge \neg(\forall xpx)$.

$(x + z) = (y \times z)$, $((x + y) \times 9) < 9$.

$\forall y((y \times ((y \times y) + y)) < y)$, or $(9 = 9) \leftrightarrow \exists z(z = 9)$.

Remark. The sets $\mathcal{T}, \mathcal{F}$ of all terms and wffs respectively of *FOL* are countable. The proof is exactly in the lines of the one establishing that the set of all PL-wffs is countable: show that the terms (and wffs) can be generated from a finite alphabet.

Let us now point out different ways in which variables may occur in a wff, in the context of the positions of quantifiers in the wff. This will prove to be critical for the FOL-semantics that we shall define in the next section.

Definition 1.3.

- (i) The *scope* of the quantifier $\forall$ in a wff $\forall x\alpha$ is defined to be α . The scope of $\exists$ in $\exists x\alpha$ is also defined to be α .
- (ii) An *occurrence* of a variable x is *bound* in a wff α if, it is immediately preceded by a quantifier, or it lies within the scope of a quantifier that is immediately followed by x .

For instance, both the occurrences of x in $\forall xpx$ are bound.

- (iii) Otherwise, the *occurrence is free* in α .

Example: consider the wff $\forall xpx \rightarrow \exists yqxy$, where p is a unary predicate symbol in the language and q , binary. The first two occurrences of x in the wff are bound, while the third is free.

- (iv) A *variable is free* in α , if it has at least one occurrence free in α . It is bound in α , if it has at least one occurrence bound in α .

If α has free variables from the list $x_1, \dots, x_n$, we write $\alpha(x_1, \dots, x_n)$ in place of α .

Clearly, a variable may be both free and bound in a wff.

(v) $\alpha(t/x)$ denotes the wff obtained from α when *all free occurrences* of x in α are replaced by the term t .

Example: consider the term $t := fyfcy$, where f is a binary function symbol and c a constant symbol in the language. Take the wff $\alpha := \forall xpx \rightarrow \exists yqxy$. Then $\alpha(t/x) := \forall xpx \rightarrow \exists yqfyfcyy$.

(vi) The term t is *free for the variable x in α* , if no occurrence of a variable of t becomes bound when any free occurrence of x in α is replaced by t .

Example: Take t and α as in (v) above. t is not free for x in α . However, if we take $t := fxx$ instead, $\alpha(t/x) := \forall xpx \rightarrow \exists yqfxxxy$, and this t is free for x in α .

(vii) A wff is said to be *closed*, if it has no free variables.

For instance, $(9 = 9) \leftrightarrow \exists z(z = 9)$ is closed.

We observe the following, using vacuous arguments.

- (a) A variable is free for itself in any wff.
- (b) A closed term is always free for any variable in any wff.

2 The semantics

In the previous section, we mentioned, informally, the intended interpretations for the symbols in an FOL-language. Let us now present the FOL-semantics formally.

Definition 2.1. An *interpretation* $\mathcal{I}$ of FOL consists of a non-empty domain D of discourse, and assignments mapping

- (i) each constant symbol c in the alphabet to a distinguished element $c_{\mathcal{I}}$ in D ,
- (ii) each function symbol f_i^n of arity n to an n -ary function $F_{\mathcal{I}} : D^n \rightarrow D$,
and
- (iii) each predicate symbol p_i^n to an n -ary relation $R_{\mathcal{I}} \subseteq D^n$.

Let us look at a couple of examples of interpretations.

Example 2.1.

- (a) Consider the FOL-language whose alphabet simply contains variables and a unary predicate symbol p . An interpretation $\mathcal{I}$ with domain $D := \mathcal{N}$, the set of natural numbers, could assign to the unary predicate symbol p , the unary relation E on $\mathcal{N}$ (a subset of $\mathcal{N}$) such that $E := \{n \in \mathcal{N} : n \text{ is even}\}$.
- (b) For another language with two binary predicate symbols, two binary function symbols and one constant symbol, we could take an interpretation $\mathcal{I}$ with the same domain, $\mathcal{N}$, and make assignments as follows: the natural number 9 to the constant symbol, the binary relations $<$ and $=$ on $\mathcal{N}$ to the two binary predicate symbols, and the binary operations $+$, $\times$ on $\mathcal{N}$ to the two binary function symbols.

We next specify assignments for terms of the language, in a given interpretation $\mathcal{I}$. Recall that $\mathcal{T}$ denotes the set of all terms.

Definition 2.2. A map $v : \mathcal{T} \rightarrow D$ is said to be a *valuation* on interpretation $\mathcal{I}$, when

- (i) $v(c) := c_{\mathcal{I}}$ for any constant symbol c in $\mathcal{T}$, and
- (ii) for a term $f_i^n t_1 \dots t_n$ in $\mathcal{T}$, $v(f_i^n t_1 \dots t_n) := F_{\mathcal{I}}(v(t_1), \dots, v(t_n))$, where $v(t_i)$, $i = 1, \dots, n$, are known.

Example 2.2. Consider the interpretation $\mathcal{I}$ in Example 2.1(b). An example of a valuation v on $\mathcal{I}$: define $v(x) := 5$, $v(y) := 1000$, $v(z) := 2$.

Then using Definition 2.2(ii) for the term $(x + y) \times 9$, we have $v((x + y) \times 9) := (v(x) + v(y)) \times 9 = (5 + 1000) \times 9$.

Note that we may have different valuations on the same interpretation, but Definition 2.2 forces all of them to assign the same value to all constant symbols, and as a result, to all the closed terms (see Assignment). So they can only differ on variables of the language.

2.1 Satisfaction, truth, validity

We now proceed to the definitions of (i) *satisfaction* of a wff by a valuation on an FOL-interpretation, (ii) *truth* and *falsity* of a wff in an interpretation, and finally, (iii) *validity* of a wff. Before we get to these, we need to look at some special kinds of valuations.

Definition 2.3. Valuations v, v' on an interpretation $\mathcal{I}$ are said to be *x-equivalent* to each other for a variable x , provided $v(y) = v'(y)$, for all variables $y \neq x$.

It is immediate that every valuation is *x-equivalent* to itself, for every variable x .

Let $\mathcal{I}$ be an FOL-interpretation as given in Definition 2.1, v a valuation on $\mathcal{I}$, and α a wff. “ v satisfies α in $\mathcal{I}$ ” is written in notation as $\mathcal{I}, v \models \alpha$. The definition of satisfaction is recursive, considering the number of connectives and quantifiers in the wff α .

Definition 2.4.

- (i) α is atomic, i.e. $\alpha := p^n t_1 \dots t_n$:
 $\mathcal{I}, v \models \alpha$, if and only if $(v(t_1), \dots, v(t_n)) \in R_{\mathcal{I}}$.
- (ii) $\alpha := \neg\beta$:
 $\mathcal{I}, v \models \alpha$, if and only if $\mathcal{I}, v \not\models \beta$.
- (iii) $\alpha := \beta \rightarrow \gamma$:
 $\mathcal{I}, v \models \alpha$, if and only if whenever $\mathcal{I}, v \models \beta$, we have $\mathcal{I}, v \models \gamma$.
- (iv) $\alpha := \forall x\beta$:
 $\mathcal{I}, v \models \alpha$, if and only if for each valuation v' that is *x-equivalent* to v , $\mathcal{I}, v' \models \beta$.

Remark.

- (a) If $\alpha := \exists x\beta$, $\mathcal{I}, v \models \alpha$, if and only if *there exists* a valuation v' that is *x-equivalent* to v such that $\mathcal{I}, v' \models \beta$. This is easily proved using the definition.
- (b) (iv) in Definition 2.4 essentially means that the wff β is satisfied by every possible value that the variable x may take in the domain (values to all other variables in the wff being those assigned by v) – in other words, by *every* element of the domain. By the same token, for satisfaction of $\exists x\beta$, we need *some* element of the domain to satisfy β .

Definition 2.5. Let $\mathcal{I}$ be any interpretation, and $\Gamma \cup \{\alpha\}$ any set of wffs.

- (i) α is *true* in $\mathcal{I}$, written $\mathcal{I} \models \alpha$, if and only if $\mathcal{I}, v \models \alpha$, for all valuations v in $\mathcal{I}$.

- (ii) α is *false* in $\mathcal{I}$, if and only if there is no valuation v such that $\mathcal{I}, v \models \alpha$.
- (iii) If every wff of Γ is true in $\mathcal{I}$, $\mathcal{I}$ is called a *model* for Γ .
- (iv) α is a *semantic consequence* of Γ , written $\Gamma \models \alpha$, if and only if every model for Γ is a model for α . If Γ is empty, α is said to be *valid*, written $\models \alpha$.

We shall see (in exercises) that the values assigned to the *free* variables by a valuation, determine whether the wff in question is satisfied by it or not. Moreover, if a wff is *closed*, it is either satisfied by all valuations, or by none.

3 Deduction in FOL

Axioms:

- A1 $\alpha \rightarrow (\beta \rightarrow \alpha)$.
- A2 $(\alpha \rightarrow (\beta \rightarrow \gamma)) \rightarrow ((\alpha \rightarrow \beta) \rightarrow (\alpha \rightarrow \gamma))$.
- A3 $(\neg\beta \rightarrow \neg\alpha) \rightarrow (\alpha \rightarrow \beta)$.
- A4 $\forall x(\alpha \rightarrow \beta) \rightarrow (\forall x\alpha \rightarrow \forall x\beta)$.
- A5 $\forall x\alpha \rightarrow \alpha(t/x)$, if t is free for x in α .
- A6 $\alpha \rightarrow \forall x\alpha$, if x is not free in α .

Ax Gen: If α is an axiom, and x is free in α then $\forall x\alpha$ is also an axiom.

Observe the role of Ax Gen: as we ‘close’ each free variable in an axiom, we get a new axiom from it; finally, the closure of the axiom (cf. Exercise ?? (i)) also gets to be an axiom.

Rule of Inference:

Modus Ponens (MP)

$$\frac{\alpha \rightarrow \beta}{\beta}$$

The Syntactic Consequence relation $\vdash$:

Let Γ be any set of wffs and α any wff in *FOL*.

Definition 3.1.

- (i) $\Gamma \vdash \alpha$, if and only if there is a sequence $\alpha_1, \dots, \alpha_n (:= \alpha)$ such that each $\alpha_i (i = 1, \dots, n)$ is either (a) an axiom, or (b) a member of Γ , or (c) derived from some of $\alpha_1, \dots, \alpha_{i-1}$ by *MP*.
- (ii) If Γ is empty in the above, we write $\vdash \alpha$, and say that α is a *theorem*, the sequence $\alpha_1, \dots, \alpha_n (:= \alpha)$ constituting a *proof* of α .

As observed for PC, we have

Proposition 3.1. $\vdash$ satisfies the following properties:

- (i) *Overlap*: if $\alpha \in \Gamma$, then $\Gamma \vdash \alpha$,
- (ii) *Dilution*: if $\Gamma \subseteq \Delta$ and $\Gamma \vdash \alpha$, then $\Delta \vdash \alpha$,
- (iii) *Cut*: if $\Delta \vdash \gamma$ for each $\gamma \in \Gamma$ and $\Gamma \vdash \alpha$, then $\Delta \vdash \alpha$.
- (iv) *Compactness*: If $\Gamma \vdash \alpha$, there is a *finite* subset Γ' of Γ such that $\Gamma' \vdash \alpha$.

The deduction theorem (D.T.) continues to hold, with identical proof as in PC.

Theorem 3.2. (Deduction) For any Γ, α, β , if $\Gamma \cup \{\alpha\} \vdash \beta$ then $\Gamma \vdash \alpha \rightarrow \beta$.

It should be mentioned here that there are other equivalent axiomatizations for FOL (e.g. refer [?]). For instance, axioms A4, A6 and Ax Gen are replaced by the single axiom:

$$\forall x(\alpha \rightarrow \beta) \rightarrow (\alpha \rightarrow \forall x\beta), \text{ whenever } x \text{ is not free in } \alpha,$$

and along with the rule MP, the *generalization* rule is considered:

$$\frac{\alpha}{\forall x\alpha}$$

Even though this alternate axiomatization looks simpler, it comes with a price: the Deduction Theorem cannot be applied as stated above. We get $\Gamma \vdash \alpha \rightarrow \beta$, when a restriction is imposed on the deduction of β from $\Gamma \cup \{\alpha\}$, namely that there should be no application of the above generalization rule, to any variable occurring free in α .

Note that in the system adopted here, the axiom above becomes a *theorem* (*Exercise*). As all of PC is kept intact inside FOL, it is no surprise that we have the following.

Proposition 3.3.

- (i) All the theorems of PC are theorems in FOL, for FOL-wffs.
- (ii) All PC metatheorems hold for FOL-wffs.

Proposition 3.4. (Equivalence theorem) If α is a wff, β a subformula of α , γ a wff such that $\vdash \gamma \leftrightarrow \beta$, and α' the result of replacing some occurrences of β in α by γ , then $\vdash \alpha \leftrightarrow \alpha'$.

Theorem 3.5. (Soundness) If $\Gamma \vdash \alpha$ then $\Gamma \models \alpha$.

Proof. The proof is by induction on the number of steps of derivation of α from Γ . In essence, one shows that (i) the axioms are valid, and (ii) *MP* ‘preserves truth’. $\square$

Proposition 3.6. (Change of bound variables)

$$\vdash \forall x\alpha(x) \leftrightarrow \forall y\alpha(y/x),$$

provided (i) y is free for x in α , and (ii) y is not free in $\alpha(x)$.

Observe that Proposition 3.6 holds, in particular, for any variable y that *does not occur* in α , as conditions (i) and (ii) are both vacuously satisfied for such a y .

4 The completeness theorem

We now proceed, much in the same lines as for PC, towards proving the completeness of FOL with respect to the semantics defined in Section 2. As we have enriched the language substantially, there is more work involved. Completeness is proved for a set $\Gamma \cup \{\alpha\}$ of *closed* wffs. Briefly put, we begin with the notion of consistency, extend it to maximal consistency; extend the language by adding constants and introduce the notion of ‘Henkin-closed’ sets; prove that (a) consistency is preserved in the extended language, (b) any consistent set in the extended language can be extended to a maximally consistent and Henkin-closed set, and (c) any maximally consistent and Henkin-closed set of closed wffs must have a model. As any model of an extension of a set also serves as a model of the set itself, (a)-(c) lead to

Theorem 4.1. If Γ is a consistent set of closed wffs, it has a model.

Theorem 4.1 leads to the completeness theorem (same proof as in PC).

Theorem 4.2. (Completeness) For a set Γ of closed wffs and a closed wff α , if $\Gamma \models \alpha$ then $\Gamma \vdash \alpha$.

Corollary 4.3. For any wff α , if $\models \alpha$ then $\vdash \alpha$.

4.1 Consistency

The same definitions as in PC are taken here.

Definition 4.1.

- (i) Γ is said to be *negation consistent* if and only if $\Gamma \not\vdash \Box$ in FOL.
- (ii) Γ is said to be *absolutely consistent* if and only if there is some FOL-wff β such that $\Gamma \not\vdash \beta$.

As in PC, a set Γ of FOL-wffs is negation consistent if and only if it is absolutely consistent, and we have

Proposition 4.4.

- (i) The set of theorems of FOL is consistent.

- (ii) If Γ has a model, it is consistent.
- (iii) Γ is consistent, if and only if every finite subset of Γ is consistent.

Definition 4.2. A set Γ of FOL-wffs is *maximally consistent*, if and only if (i) it is consistent, and (ii) $\Gamma \cup \{\alpha\}$ is *inconsistent*, whenever $\alpha \notin \Gamma$.

Proposition 4.5. Let Γ be maximally consistent, α, β any wffs in FOL.

- (i) $\Gamma \vdash \alpha$, if and only if $\alpha \in \Gamma$.
- (ii) $\neg\alpha \in \Gamma$ if and only if $\alpha \notin \Gamma$.
- (iii) $\alpha \rightarrow \beta \in \Gamma$ if and only if $\alpha \notin \Gamma$, or $\beta \in \Gamma$.
- (iv) $\alpha \wedge \beta \in \Gamma$ if and only if $\alpha \in \Gamma$ and $\beta \in \Gamma$.
- (v) $\alpha \vee \beta \in \Gamma$ if and only if $\alpha \in \Gamma$ or $\beta \in \Gamma$.

Note. Henceforth, we assume that Γ is a set of closed wffs and α is a closed wff.

Proposition 4.6. If Γ is consistent, it has a maximally consistent extension.

4.2 Henkin-closed sets

As we mentioned in the beginning of this section, we require a little more work in FOL, in order to arrive at a model for a consistent set. We introduce the notion of a ‘Henkin-closed’ set as follows.

Definition 4.3. A set Γ is said to be *Henkin-closed* or simply, *H-closed*, provided the following hold.

- (i) The language of Γ has at least one *closed term* (i.e. a term with no variable).
- (ii) For any closed wff $\forall x\alpha$ in the language, we have $\Gamma \vdash \forall x\alpha$, whenever $\Gamma \vdash \alpha(t/x)$ for all closed terms t .

It can be proved that any consistent set can be extended to a maximally consistent *and* H-closed set, and that every maximally consistent and H-closed set must have a model. For this purpose, we need to extend the given FOL language $\mathcal{L}$ to include a *countably infinite set of new constants* giving a larger language $\mathcal{L}'$.

Proposition 4.7. Any consistent set can be extended to a maximally consistent and H-closed set in $\mathcal{L}'$.

Proposition 4.8. If Γ is maximally consistent and H-closed, it has a model.

Proof. Let Γ be maximally consistent and H-closed. Define the canonical interpretation $\mathcal{I}$ as follows.

- The domain D of $\mathcal{I}$ is the set of *all closed terms* of the language $\mathcal{L}'$.
- If p^n is a predicate symbol of arity n , it is mapped to an n -place relation R on D , defined as:

$$(t_1, \dots, t_n) \in R, \text{ if and only if } \Gamma \vdash p^n t_1 \dots t_n,$$

for any members $t_1, \dots, t_n$ of D .

- If f^n is a function symbol of arity n , it is mapped to an n -ary function F on D , defined as:

$$F(t_1, \dots, t_n) := f^n t_1 \dots t_n,$$

where $t_1, \dots, t_n \in D$.

- A constant symbol c is mapped to itself.

Note that, because we are in $\mathcal{L}'$, $D \neq \emptyset$. The following would establish that $\mathcal{I}$ is a model for Γ .

Claim. For any closed wff α , $\mathcal{I} \models \alpha$, if and only if $\Gamma \vdash \alpha$.

□

Propositions 4.6 and 4.8 give us Theorem 4.1, which finally yields the completeness of FOL for closed wffs (Theorem 4.2).